

SPINAL LTP INDUCED BY SCIATIC NERVE ELECTRICAL STIMULATION ENHANCES POSTERIOR TRIANGULAR THALAMIC NOCICEPTIVE RESPONSES

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Abstract—Long-term potentiation (LTP) can be induced by electrical stimulation and gives rise to an increase in synaptic strength at the first relay. This phenomenon has been associated with learning and memory and also could be the origin of several pathological states elicited by an initial strong painful stimulus, such as some forms of neuropathic pain. We used high-frequency electrical stimulation of the sciatic nerve in anesthetized rats to produce spinal LTP. To evaluate the effect of spinal LTP on the activity of neurons in the posterior triangular nucleus of the thalamus (PoT), we applied an electrical stimulation (40 stimuli; 1 ms; 0.5 Hz; 1.5 mA) to cutaneous tissues at 10-min intervals during at least 3 h. In the majority of cases, PoT cells did not respond to cutaneous stimulation before LTP, but 50 min after LTP induction PoT cells progressively began responding to the cutaneous stimulation. Furthermore, after 3 h of LTP induction, PoT neurons could respond to cutaneous stimulation applied to different paws. Interestingly, the conduction velocities for the receptive field responses from the paw to the PoT cells were compatible with those of A δ -fibers. Since PoT cells project to the insular cortex, the progressive increase in PoT activity and also the progressive unmasking of somatic receptive fields in response to LTP, place these cells in a key position to detect pain stimuli following central sensitization. © 2013 IBRO. Published by Elsevier Ltd. All rights reserved.

Key words: long-term potentiation, hyperalgesia, poly-synaptic action.

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Abbreviations: AUC, area under curve; HFS, high frequency stimulation; LTP, long-term potentiation; PoT, posterior triangular nucleus of the thalamus; RF, receptive field; S.E.M., standard error of the mean; WDR, wide dynamic range cells.

INTRODUCTION

Long-term potentiation (LTP) elicited in central structures has been widely used as a model of synaptic plasticity, reflecting synaptic strengthening in synapses subjected to repetitive activation (Bliss and Lomo, 1970, 1973; Clugnet and LeDoux, 1990; Bliss and Collingridge, 1993). In this context it has been suggested that spinal LTP plays a key role in the development of central sensitization leading to persistent pain (Woolf, 1983; Randic et al., 1993; Liu and Sandkühler, 1995). Indeed, perceptual correlates of spinal LTP have also been reported in humans (Klein et al., 2004) and may be one of the mechanisms whereby acute pain elicits central sensitization phenomena related to hyperalgesia or allodynia (Svendsen et al., 1999a,b; Rygh et al., 2002; Sandkühler, 2009; Sandkühler and Gruber-Schoffnegger, 2012).

Spinal nociceptive processing, including spinal LTP, is controlled by a number of descending modulatory systems (Svendsen et al., 1999a,b; Gjerstad et al., 2001; DeLaTorre et al., 2009; Condés-Lara et al., 2012). Spinal LTP could be at the origin of chronic pains that develop following an initial painful event (Ruscheweyh et al., 2011). However, little is known about how spinal LTP could affect thalamo-cortical nociceptive networks. We aimed to study the effects of spinal LTP on neurons of the thalamic, triangular posterior nucleus (PoT) since the PoT is one of the main targets of spinal and trigeminal lamina I neurons (Gauriau and Bernard, 2004a). Moreover, PoT neurons project to the posterior insular cortex (ICx), a structure (Gauriau and Bernard, 2004b) that is known to be involved in pain perception in humans (Isnard et al., 2011; Garcia-Larrea, 2012a,b) and nociceptive processing in rodents (Coffeen et al., 2011; Benison et al., 2011). The aim of the present study was to investigate the effects of spinal LTP on the activities of PoT neurons following high-frequency electrical tetanizing stimulation of the sciatic nerve (Liu and Sandkühler, 1995, 1998; Svendsen et al., 1999a,b). We found evidence that spinal LTP is able to increase the excitability of PoT neurons.

EXPERIMENTAL PROCEDURES

Animals

Male Wistar rats (260–300 g) from the Neurobiology Institute Animal House were used in the present experiments. The animals were housed in plastic cages in a special temperature-controlled room (22 $\pm$ 2 °C) on a 12-h:12-h light/

dark cycle and given *ad libitum* access to food and water. Our Institutional Ethics Committee approved all animal procedures, and in the experimental protocols we followed the IASP ethical guidelines (Zimmermann, 1983) and the guidelines contained in the NIH guide for the Care and Use of Laboratory animals (80-23, revised in 1996). All efforts were made to limit distress and to use only the number of animals necessary to produce reliable scientific data.

General methods

Surgical procedures. Thirty rats were initially anesthetized in a hermetic box with sevoflurane (3%) in a mixture of two-thirds N₂O and one-third O₂. An intra-tracheal cannula was inserted for artificial ventilation and to maintain the anesthesia throughout the experiment; then, the animals were mounted on a stereotaxic frame. All the surgical procedures (see below) were made using 3% sevoflurane. The subsequent experimental protocols (see “Experimental protocols”) were performed under 2–2.5% sevoflurane in order to achieve the level of anesthesia that was adequate for ethical purposes but without excessively depressing neuronal responses to noxious stimuli (DeLaTorre et al., 2009). As previously reported, the animals were not paralyzed, and we did not observe a withdrawal reaction during the experiments (Condés-Lara et al., 2009, 2012). End tidal CO₂ was monitored with a Capstar-100 CO₂ analyzer (CWI Inc., Ardmore, PA, USA) and kept within the physiological range (2.5–3.0%) throughout the experiment. Core body temperature was maintained at 38 °C using a circulating water pad.

The left sciatic nerve was carefully exposed and isolated at the mid-thigh level. A pair of silver hook electrodes was placed under the sciatic nerve for electrical stimulation (Wire, Silver, Teflon, 0.10 Bare 0.13 Coated, PHYPEP s.a.r.l. Paris), and the exposed nerve was covered with mineral oil at 37 °C. Finally, a craniotomy (~2 × 4 mm) contralateral to the sciatic nerve surgery was performed in the region overlaying the lateral thalamus. The dura mater was removed from the exposed region of cortex to allow access to the PoT.

Extracellular PoT unitary recordings. Using the stereotaxic coordinates (AP 3.70; L 2.50; H 6.0 ± 0.4) of Paxinos and Watson (1998), extracellular unitary recordings were made with: (i) a glass micropipette (impedance 4–8 MΩ) filled with 3% Pontamine Sky Blue (1 M KCl solution) (12 rats) or (ii) seven quartz–platinum/tungsten microelectrodes (impedance 4–7 MΩ) mounted in a multichannel microdrive “System Eckhorn” (Thomas Recording, Giessen, Germany) (18 rats). Due to the low electrode impedances with the Thomas Recordings we could not obtain single-spike recordings, but the multi-unit activities (2 or 3 different spikes) recorded were processed with the spike 2 software to identify single units based on amplitude and shape. The same procedure allowed us to distinguish between neuronal activities and stimulus artifacts. However, we only reported single units that, in the case of multi-unit activities, we were able to follow throughout the entire experiment (more than 180 min).

Fig. 1 illustrates a recording in the control situation, before the high-frequency stimulation (HFS) of the sciatic nerve and after 120 min of sciatic HFS to induce LTP (c.f., section “Protocol II: induction of spinal LTP by sciatic nerve electrical stimulation”). Fig. 1 also illustrates the use of digitized filters from the spike 2 software that allowed us to distinguish between the single unit potentials and stimulus artifacts. Fig. 1B shows a single digitized response at min 120 after sciatic HFS. This type of response was not observed before the HFS, and its appearance started after at least 50 min of HFS. Fig. 2 illustrates three responses, and the corresponding digitized signals, to receptive field (RF) stimulation at 180 min

after HFS. The digitization process was conducted with great care to separate single-unit activities (due to the low resistance of the electrodes) when the recordings come from multiunit activities.

The glass microelectrodes were fixed to a micromanipulator drive to reach the cells recorded from the PoT, while the multielectrode system was manipulated with the 7-channel version of Fiber-Electrode Manipulator “System Eckhorn” using Ekhn Matrix Multiuser Software (Thomas Recording, Giessen, Germany).

Vertical electrode penetrations were carried out; after entering the PoT (5600–6400 μm from the surface of the cortex) the electrode was advanced in small steps (3–6 μm). During the exploration, when a single suspected PoT cell was found, we manually tested the four limbs to locate the receptive field region that could activate the cell. Note that this sampling method ignores nociceptive neurons that are silent, but it has the advantage of being more natural and thus avoids damage caused by repeated noxious stimuli (Gauriau and Bernard, 2004b).

The extracellular neuronal activity was amplified, digitized, and discriminated using CED hardware and Spike 2 software (Cambridge Electronic Design; Version 5.20). Waveforms and recorded spike trains were stored on a hard drive for off-line analysis. To identify the responding cells, action potentials of recorded cells were digitized and analyzed using peri-stimulus time histograms and raster displays (1 s duration; bin time = 1 ms). Identified PoT neurons responding to non-noxious stimuli (tactile and pressure) with a defined somatic RF were rare; for this reason we systematically applied electrical pulses (see Protocol I) on the right posterior paw to start and to have an initial signal for measuring the neuronal activity. Under these experimental conditions, when a stable recording was achieved for several minutes from which we could distinguish a single-unit potential with the appropriate amplitude and shape as well as a depth corresponding to the PoT stereotaxic coordinates, the signal was considered to come from a suitable PoT cell, and we proceeded to test the effects of spinal LTP on neuronal activity using one of the protocols below (see below). We only included cells in the analysis whose location in the PoT was confirmed histologically. In the sham experiments, the sciatic nerves were carefully exposed and isolated, and a pair of silver hook electrodes was placed on them, but they were never stimulated.

Experimental protocols

Protocol I: electrical stimulation of the somatic receptive field. Using two fine needles (attached to a stimulus isolator unit) inserted subcutaneously into the right posterior paw, electrical square-wave stimuli test (see below) was delivered (Grass S88 Stimulator): (i) 3 or 4 times at 10-min intervals (in order to obtain a control before sciatic HFS to induce spinal LTP as it was described in the section “Protocol II: induction of spinal LTP by sciatic nerve electrical stimulation”); (ii) immediately after the HFS at the sciatic nerve as a conditioning stimuli ($t = 0$); and (iii) every 10 min for 1 h and every 20 min for the next 4 h after spinal HFS to LTP induction. Electrical stimulation of the right posterior paw consisted of a train of 40 stimuli at 0.5 Hz, with 1-ms pulse duration at 1.5 mA. This intensity of somatic stimulation was considered*** as noxious somatic stimulation, because in previously reported experiments using lower intensities we evoked Aδ and C fiber activation when recording spinal dorsal horn cells (Condés-Lara et al., 2005, 2009, 2012; DeLaTorre et al., 2009). These procedures allowed us to compare neuronal responses of the recorded cells before and after spinal LTP induction.

Protocol II: induction of spinal LTP by sciatic nerve electrical stimulation. Spinal LTP (Sandkühler, 2007) was elicited by sciatic nerve stimulation with high-frequency stimulation (HFS)

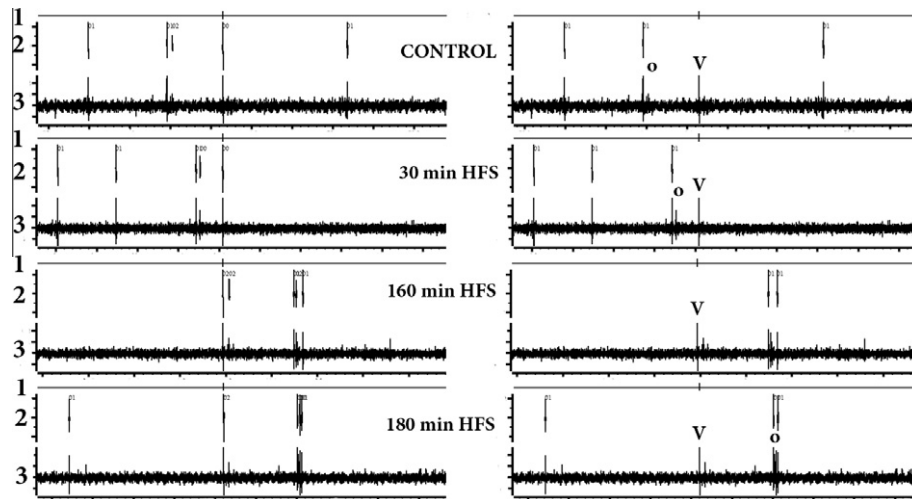


Fig. 1. Procedure to distinguish single-unit from multi-unit activities or stimulus artifacts in the raw data, in the control, 30, 160 and 180 min after HFS using the Spike 2 digital filters. Traces labeled as 1 represent stimulus artifact, 2 digitized signals and 3 the raw data recordings. The right and left columns of recordings are the same but in the right columns (in traces 2) only the neuronal activities coded as 01 were plotted, which correspond to the single cell followed during the entire experiment. In the right column the (Y) corresponds to the suppressed stimulus artifact by the Spike 2 digital filters and the (o) corresponds to the suppression of action potentials different to the single neuronal activities by using digital filters. Different action potentials were considered when the shape and amplitude of actions potentials differed by 20%. Note that with this filter system, we only analyzed, in this case, the spike coded as 01. The recording duration for each trace was 500 ms.

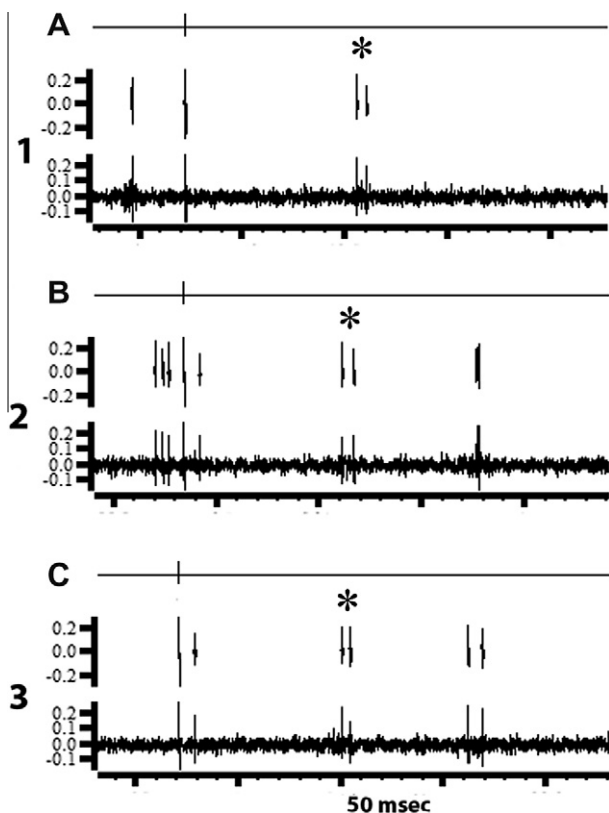


Fig. 2. Raw data of the recorded responses (*) 180 min after sciatic HFS. Three responses to noxious somatic stimulation are illustrated in panels 1, 2, and 3. Traces A–C represent the stimulus artifacts. The second traces were the digitized waveforms coming from the recorded activity illustrated in the third traces. The responses evoked by noxious receptive field stimulations were not observed before the sciatic HFS.

(100 Hz), suprathreshold intensity (300 μ A) with square-wave pulses of 1-ms duration; seven stimulating trains with 6 s on

and 6 s off were delivered (Grass S88 Stimulator). The intensity used produced a significant and sustained paw extension for the full duration of the stimulus. In the sham group, the exposed left sciatic nerve was placed on the stimulation electrode, but was not stimulated.

Histology and reconstruction of recording sites

At the end of the recording track, the position of the cells recorded with glass microelectrodes was marked by iontophoretic injection of Pontamine Sky Blue (cathodic current 10–15 μ A, 30 min). The animals received an overdose of anesthesia (80 mg/kg of pentobarbital, until cardiac arrest), and then they were immediately perfused transcardially with saline solution followed by 10% formalin (~200 ml of each). Brains were postfixed in 10% formalin and cut in serial coronal sections of 40- μ m thicknesses with a Leica SM 200 R freezing microtome (Leica Instruments BrmbH, Nussloch, Germany). Recording sites were observed with a light microscope and confirmed by the depth measurements, and the histological track locations were plotted schematically. In the case of multiple microelectrode recording with the Thomas apparatus, a careful histological analysis allowed us to identify at least 5–6 electrode tracts. In Fig. 3 we illustrate the histological reconstruction of the electrode tracts and the final recording points.

Data analysis

Our results are presented as mean $\pm$ standard error of the mean (S.E.M.). The neuronal firing rates during 200 ms before and 800 ms after electrical stimulation of the right posterior paw, were compared by using a two-way repeated measures analysis of variance. The data collected were expressed as events by seconds (Hz) as a normalization procedure (Fig. 4). In order to compare the change of the firing rate induced by electrical stimulation of the right posterior paw in sham or LTP animals, the number of evoked potentials was expressed as percent change from the respective baseline (100%, as it is illustrated in Fig. 6). Note that the baseline value refers to the

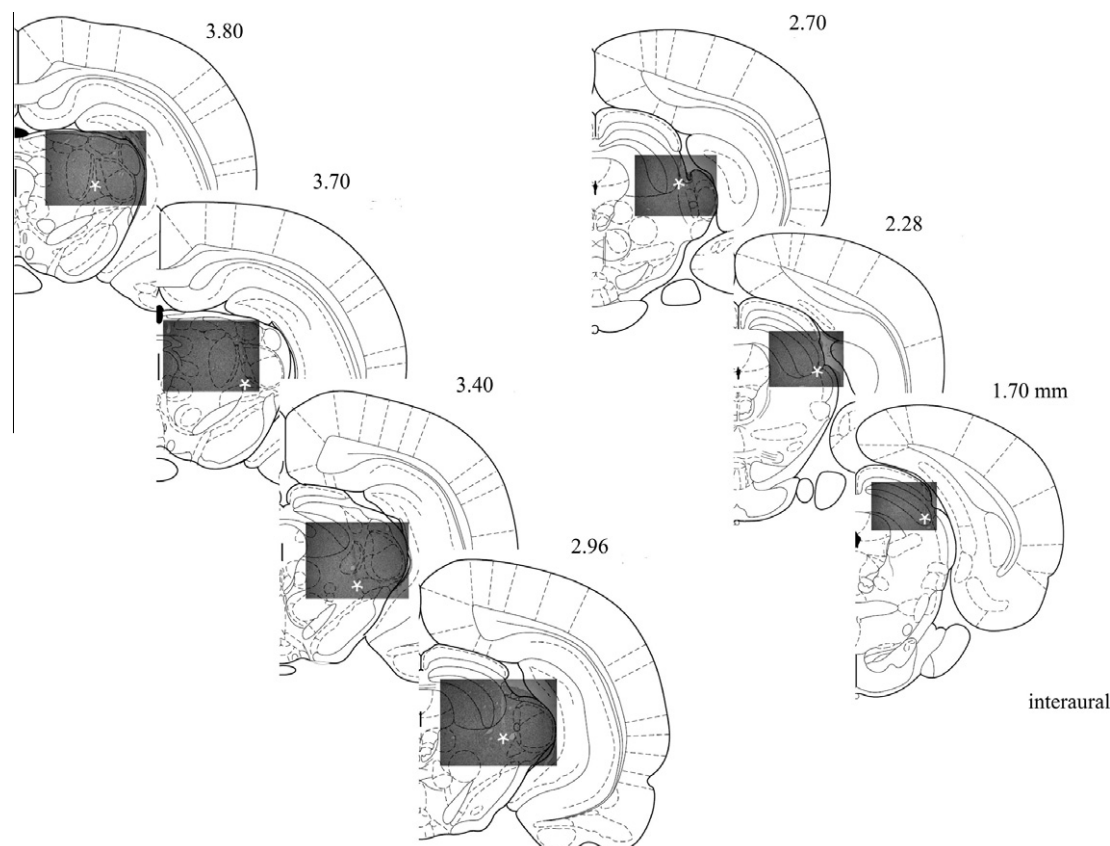


Fig. 3. Location of the recording sites with the Thomas system. The trajectory and tip locations are illustrated in the photomicrographs superimposed on each diagram. Only electrodes located at 3.80, 3.70, and 3.40 were found in the PoT. The exact location of the recorded electrodes (*) was adjusted using the reference microdrive records. The anterior posterior planes follow interaural reference points from the Paxinos atlas.

evoked neuronal response before spinal LTP. The difference expressed as percent of change in the neuronal activity evoked within one group of animals before and after spinal LTP induction, was compared by using a one-way repeated measures analysis of variance. Finally, the areas under curve (AUC) between the sham and spinal LTP groups were compared using the Student's *t*-test. The one- and two-way repeated analysis of variance was followed, if applicable, by the Student–Newman–Keul's post hoc test. The threshold for statistical significance was set at $P < 0.05$. It is important to mention that the values of Hz and percent of change obtained in the present experiments correspond to the mean $\pm$ S.E.M for the 40 stimuli applied at the receptive field.

RESULTS

General properties of recorded cells

We studied a total of 37 cells; the position within the PoT was confirmed histologically for 15 of these cells (from nine rats). These 15 cells were recorded over a period of at least 3 h and did not exhibit a defined RF for natural (tactile and pressure) or cutaneous electrical stimulation in control or sham experiments. Nine cells were studied after inducing LTP using HFS, and the other six were studied as sham cells without HFS. The mean depth of the recorded cells was $5300 \pm 200 \mu\text{m}$. The other 22 cells did not present changes associated with the receptive field stimulation as a consequence of

the sciatic HFS, and all of them were found to be outside of the PoT.

Responses evoked by electrical stimulation

As described above, we were unable to obtain a consistent evoked response associated with a specific RF. For this reason we choose to make the arbitrary electrical stimulation in the right posterior paw since we could observe a general increased firing rate after stimulation of the paw (see Fig. 4). This increase in the firing rate following the cutaneous stimulation led us to refer to the right posterior paw as the RF (see control situation before and after LTP induction) and then to study the effects of spinal HFS upon the evoked activities in PoT cells.

After sciatic HFS to induce spinal LTP, we also found a significant increase in the neuronal firing rate when the activity evoked by cutaneous electrical stimulation was compared before and after the RF stimulation (Fig. 4). These progressive increases in the neuronal activity were more pronounced at 180 min. In Fig. 5, a raster display of the digitized responses (an example is illustrated in Fig. 2) shows that after spinal LTP induction, the responses evoked by electrical stimulation of the RF progressively increased beginning 50 min

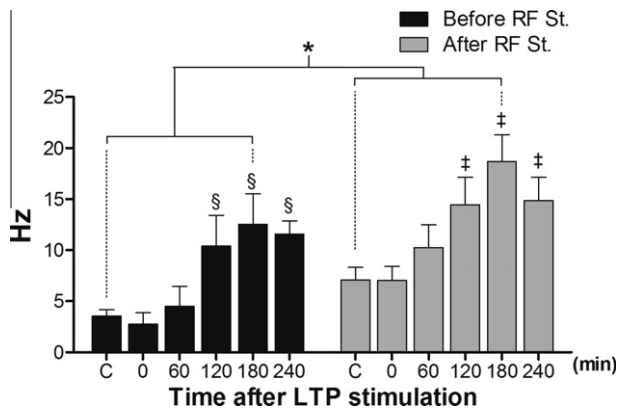


Fig. 4. Firing rate of PoT cells. We compared the activity 200 ms before (black columns) and 800 ms after (gray columns) the 40 noxious receptive field stimulation in C (control = before HFS), and at 0, 60, 120, 180, and 240 min after the sciatic HFS. The firing rate activities (expressed as spikes per second, Hz) before and after RF stimulation were different (keep in parentheses: * $P < 0.05$ when comparing the firing rate at C, 0, 60, 120, and 180 min; § $P < 0.05$ when comparing the activity in the control situation (C) vs that 120, 180, and 240 min before the RF stimulation; ‡ $P < 0.05$ when comparing the activity in C vs that 120, 180, and 240 min after the RF stimulation). Notice that although the firing rate is enhanced by the RF stimulation, the spinal LTP potentiated this effect. The comparison in the control situation (before the sciatic HFS) between before and after RF noxious stimulation indicates a general increase in the firing rate but without the time-locked responses shown in Figs. 1B and 2A firing rate increase due to the RF stimulation before the HFS was one of the criteria for selecting the cells to be studied.

after HFS and expanded to other “new” receptive fields located in the anterior paws.

HFS activation and RF evoked responses

HFS of the sciatic nerve was used to induce LTP and this, in turn, produced a long-lasting post-discharge of a few minutes in PoT cells. Spinal LTP induction progressively increased the neuronal activity induced by electrical stimulation in identified PoT neurons, and this was significantly different when compared with non-sciatic HFS animals or sham experiments (Fig. 6). This increased activity became significant 100 min after LTP induction. Importantly, the PoT cells did not respond in a time-locked way (PoT cells only increased their firing rate as a global response, as illustrated in Fig. 4) to a particular cutaneous region prior to the HFS, but after HFS of the sciatic nerve the PoT cells started to display clear activity evoked by the cutaneous RF electrical stimulation.

Fig. 7 illustrates the progressive evolution of activity associated with the RF stimulation detectable after 10 min after, but more clearly 50–100 min after the LTP induction, when a clear RF-evoked response with a latency of 82.5 ± 5 ms and duration of 50 ms was observed. Also, as a part of the evoked response, a less prominent increase in the neuronal firing rate was observed for 300 ± 16 ms in the peristimulus time histogram illustrated in Fig. 7A.

Moreover, trains of high-frequency multispikes (only presented in HFS rats) continued throughout the entire experiment. These trains of multispikes produced by

HFS could also be observed in the interval between the successive trains of sciatic stimulation, presenting a characteristic pattern of multispikes discharges lasting 50 ± 5 ms. and they became progressively associated with RF electrical stimulation responses. These high-frequency multispikes are illustrated in Figs. 7 and 8.

Using the Thomas system with seven electrodes, in one experiment we simultaneously recorded three histologically identified PoT cells (see Fig. 3). In order to test if the repetitive stimulation of the RF was sufficient to elicit the RF-evoked responses, we carried out eight repeats at 10-min intervals of the 40, 0.5 Hz stimulations of the RF as a control. The cell responses were not affected during this period of control stimulation but later, when we used HFS to induce LTP, the cells presented an increased response to RF stimulation. Note that cell responses are not dependent on the basal activity and that they have a similar time course. These results are illustrated in Fig. 8. At the end of one experiment, the cutaneous electrical stimulation was applied to the forepaws, and we found responses similar to those produced by the stimulated RF on the right posterior paw. These results are illustrated in Fig. 5 (right p and left a).

The increased neuronal firing rate following LTP and the progressive increase of the associated RF responses (time-locked evoked activities) was not observed in the cells recorded at other locations near the PoT, or in sham experiments. Fig. 9 shows the locations of the PoT cells recorded.

DISCUSSION

This study shows that the induction of spinal LTP by high-frequency stimulation (HFS) enhances the excitability of PoT neurons and suggests that the projecting dorsal horn spinal cord cells are implicated in this pain processing amplification following peripheral conditioning stimulation. Regarding our experimental protocols, we did not measure spinal LTP directly (classically at the first synaptic relay), but the criteria used for HFS of the sciatic nerve were that it: (i) enhances synaptic transmission in the spinal cord and (ii) leads to hyperactive dorsal horn neurons (Svensen et al., 1999a; DeLaTorre et al., 2009), both of which are processes correlated with central sensitization (Woolf and Salter, 2000) and the induction of spinal LTP (Sandkühler and Gruber-Schoffnegger, 2012).

Effects of the RF stimulation on the neuronal activity of PoT neurons

Under our experimental conditions we did not observe evoked responses (time-locked activities associated with the electrical RF stimulation) following RF stimulation in control and sham experiments; however, we detected an increase in the general firing rate of PoT neurons after spinal LTP (see Fig. 6). It is important to stress that in control (before the HFS) the receptive field stimulation enhanced the neuronal firing rate as a generalized response as illustrated in the Fig. 4. However, the RF stimulation was generally associated

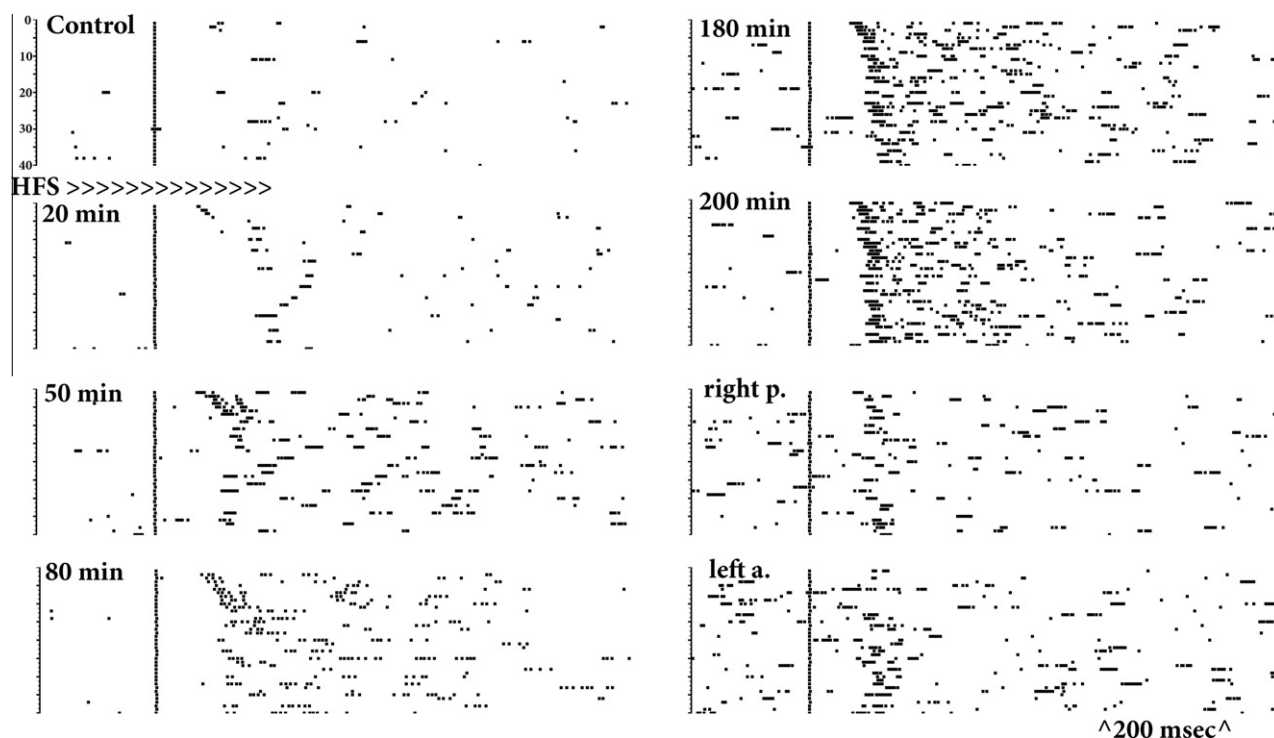


Fig. 5. Digitized action potentials were used to construct the dot raster display showing the RF activity in C, the control situation, and at 20, 50, 80, 180, and 200 min after HFS >> to induce a LTP. Also shown are the responses produced by stimulation of the right anterior paw (right p) and the left anterior paw (left a.). Notice the progressive time-locked responses beginning at 50 min and the responses produced by the anterior paw stimulations.

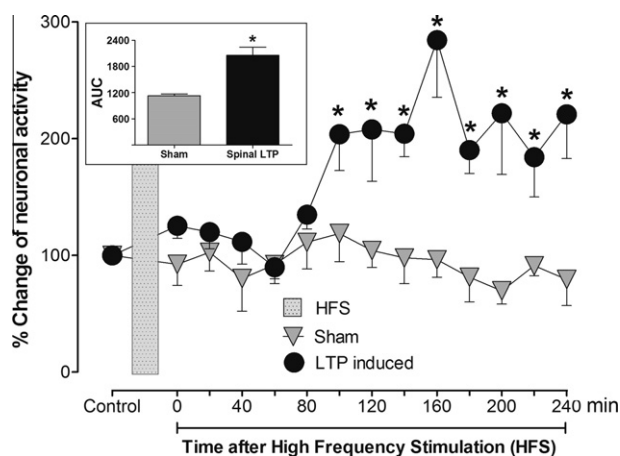


Fig. 6. Neuronal firing rate induced by the receptive field cutaneous (RF) stimulation in the control (before LTP) and each 20 min after the LTP induction. The graph compares the activity in sham experiments vs LTP experiments. The insert shows the total activity expressed as AUC in sham and LTP spinal neurons. * $P < 0.05$ vs control (C) or sham.

to time-locked response but also RF stimulation not only induced time-locked responses but it could also induce a generalized increase in activity or a rhythmic pattern of oscillation in the neuronal discharges. The spontaneous PoT firing rate was also enhanced by HFS stimulation when compared with the control, previous to the RF stimulation (see Fig. 4) and also when compared with sham experiments (Fig. 6). Interestingly, in controls

or prior to LTP induction, we could not detect any clear structured evoked responses associated with RF stimulation. Guilbaud et al. (1980) and Gauriau and Bernard (2004b) described neuronal responses in the PoT similar to wide dynamic range, non-noxious and noxious-specific cells. However, we were not able to reproduce these results in control conditions; the main methodological difference was the use of different anesthetics. Guilbaud et al. (1980) and Gauriau and Bernard (2004b) used halothane, and in our experiments we used sevoflurane in non-paralyzed animals. Initially we tried, unsuccessfully, to locate receptive fields using mechanical stimulations with different intensities, our present results were induced by electric stimulation of the skin. In contrast, Guilbaud and Gauriau used thermal and mechanical stimulations.

Effect of spinal LTP induction on the neuronal responses of PoT neurons

The spontaneous PoT firing rate was also enhanced by HFS stimulation when compared with controls (see Fig. 4). The main point to discuss is the progressive, post-LTP organization of the evoked and associated responses to RF stimulation. A major finding was that in controls or prior to LTP induction, we could not detect any organization in the evoked responses to RF, but following HFS the peristimulus histograms presented a very well-defined response with a mean latency of 82.5 ± 5 ms, lasting 50 ms (from 50 to 110 ms after RF stimulation) followed by a later postdischarge-evoked

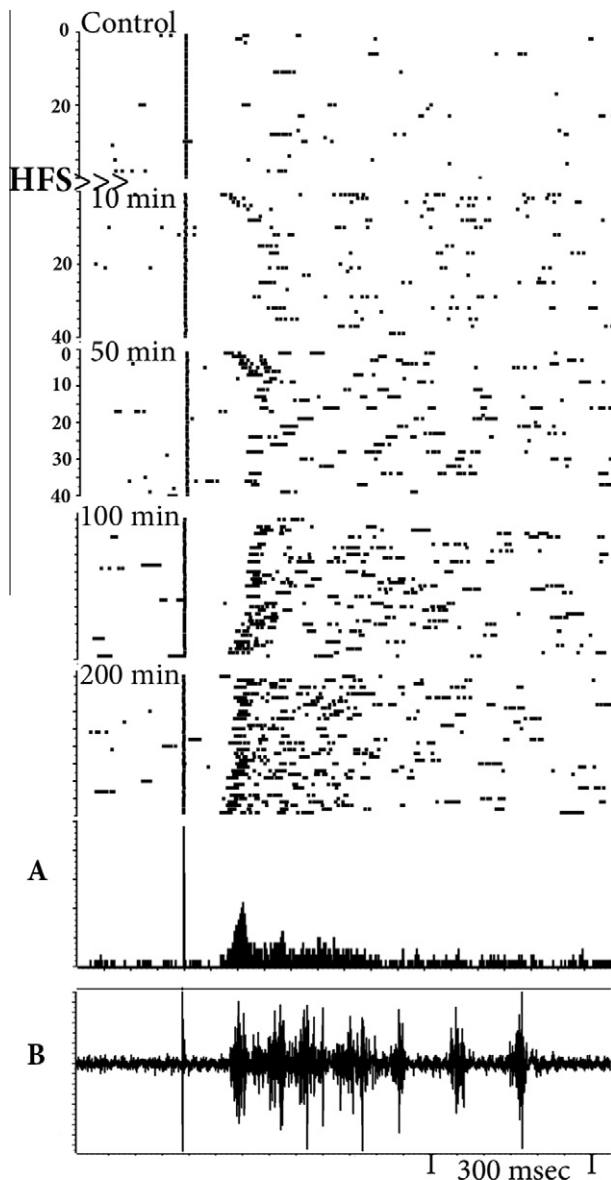


Fig. 7. Dot raster display showing the progressive increase in the general activity and the presence of the time-locked responses produced by RF stimulation after HFS > > > to LTP induction. The increases in the responses were more evident at 50, 100, and 200 min after sciatic HFS. Panel A shows a peristimulus time histogram at 200 min after LTP, and panel B illustrates the bursts of epileptic-like activity produced by RF stimulation. This type of activity was observed only after LTP induction by sciatic HFS.

response lasting for 300 ± 16 ms (see Figs. 2, 7 and 8). An important question arises as to which type of fibers mediate this response (82.5 ± 5 ms).

In the spinal cord at the L4 level, the electrical stimulation of hind limb evoked responses with latencies between 20 and 90 ms for A δ -fibers, and responses with latencies between 90 and 300 ms were considered to reflect C-fiber activation. Thus, it is likely that the evoked responses in the PoT are mediated by peripheral activation of A δ -fibers, certainly we cannot exclude the participation of C fibers or other type of fibers generating a generalized enhancement of activity

as it is illustrated in Fig. 6 when compared the neuronal firing rate after HFS with the neuronal activity in sham experiments. Previous studies demonstrating that there exists a projection from spinal lamina I neurons to PoT neurons (Gauriau and Bernard, 2004a) gives support to the idea that spinal superficial neurons are implicated in the process that increases the information flow to the PoT. In any case this response has a short duration and is due to homotopic activation at the spinal cord level. On the other hand, the postdischarge activation lasting 300 ms could be due to heterotopic activation at the spinal cord level (Ruscheweyh et al., 2011).

Development of specific RF on PoT neurons after spinal LTP: implications for pain processing

Sciatic HFS induced profound changes in the response of thalamic neurons to cutaneous stimulation. The present study is the first to demonstrate changes in the response properties of neurons in the PoT. All of the PoT neurons tested in these experiments developed a RF located in the right posterior paw and expanded this receptive field to other paws after spinal LTP (see Fig. 5). It is important that this RF location was not necessarily unique, and that it was the location that was tested before and after stimulation. The current results are of particular interest because they show that during sciatic LTP, the functional properties of thalamic neurons will markedly change due to the exacerbated information provided by spinal cells to signal further incoming noxiously stimulated receptive fields (stimulation of other body parts). So, the PoT cells acquired or unmasked a new property, detecting noxious stimulations at different body surfaces. We can only demonstrate an evolution and congregation of action potentials as a new response following several minutes of HFS. This is an intriguing point for the functional roll of PoT cells. What are they for? We suggest that the PoT is part of a system involved in the processing of pain perception, in particular for allodynia, which allows different parts of the body during noxious stimulations to be identified. This point is related to other properties of pain sensations, such as the identification, the location and the emotional compounds that have been claimed and signaled since a long time ago (Albe-Fessard et al., 1985).

In this way we need to keep in mind that an intimate relationship between superficial cells and deep WDR neurons in the spinal cord exists, as suggested by Rygh et al. (2006). Indeed, LTP of C-fibers mediates nociceptive transmission of neurons located in the superficial and deep dorsal horn (Ruscheweyh et al., 2011). Furthermore, repetitive stimulation of LTP, enhanced and unmasked the receptive noxious field on a very large fraction of the body surface. A locally expanded receptive field due to the activation of low-threshold sensory afferents by innocuous mechanical stimulation in the hind paw receptive field, evoked the typical brief excitation in control rats. However, in neuropathic rats, innocuous stimulation also induced a nociceptive-like after discharge that persisted for 2–3 min. Our data provide evidence of central effects that can explain how noxious A δ - and C-fiber activation (by LTP) is linked to central processes of sensitization. Thus, LTP

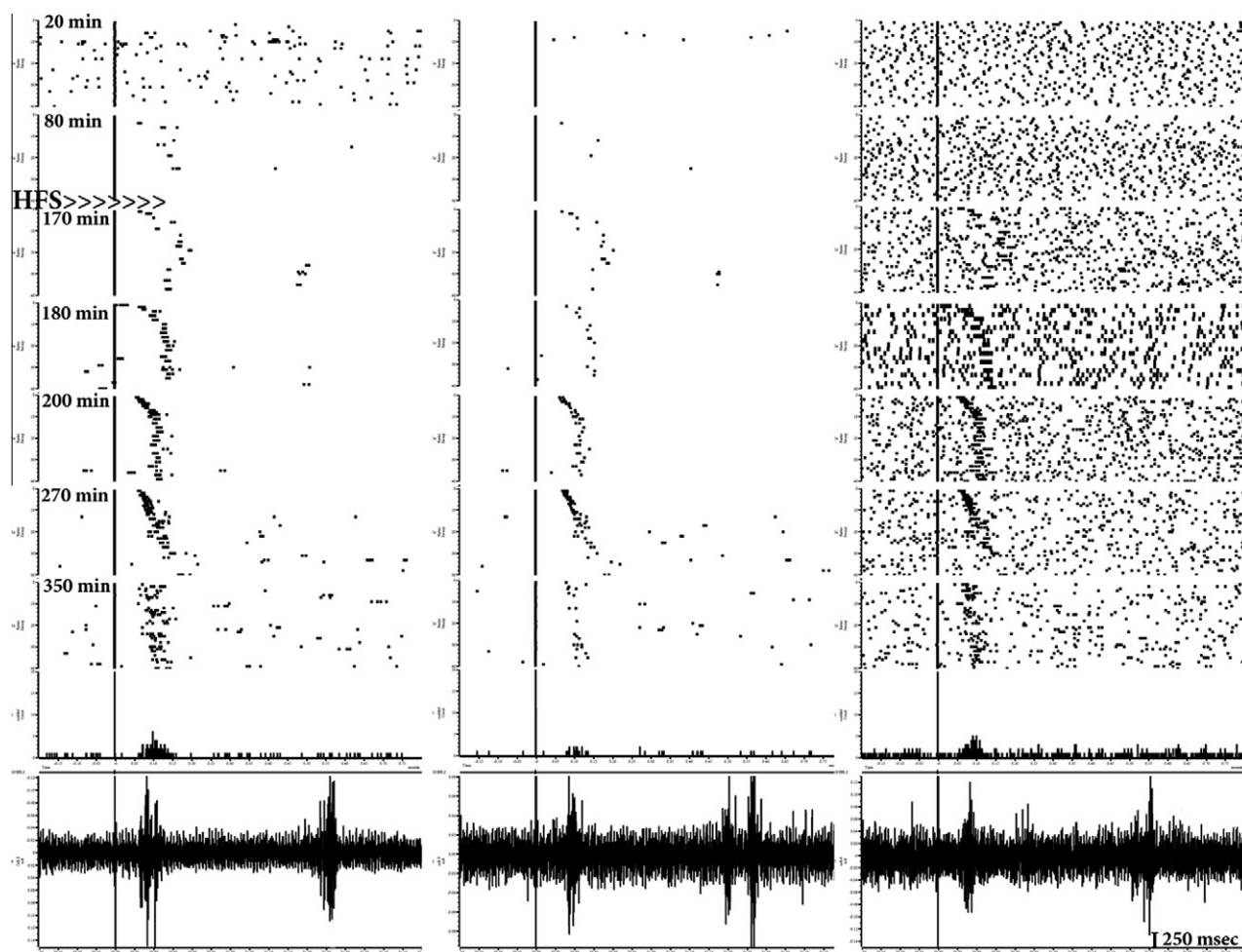


Fig. 8. Raster dot display obtained from three simultaneously recorded cells. From top to bottom: Control-evoked activity, HFS > > > > to LTP induction and raster dot display at 170, 180, 200, 270, and 350 min after LTP induction. Panels in A show the peristimulus time histograms at 350 min, and panels in B show examples of the activity evoked by RF stimulation. Notice the burst discharges in B, as responses.

could be widespread and involve thalamic structures as a general process of activation and may not be restricted to the first synaptic relay.

Final considerations

Spinal LTP, a phenomenon linked to central sensitization and consequently to the development of hyperexcitability, represents a cellular model for pain amplification (hyperalgesia) (Ji et al., 2003; Woolf, 2011), and it is possible that PoT neurons can be an important component responsible for the development of allodynia due to an increased representation of the noxious stimulation in several body regions (see Fig. 5).

Spinal LTP induced by sciatic nerve HFS could activate A δ - and C-fiber-mediated nociceptive transmission in superficial and deep dorsal horn neurons, and this is considered to be a key mechanism responsible for the increased responsiveness of dorsal horn neurons. Moreover, several studies have focused on the homosynaptic component of LTP; nevertheless, there are relevant central changes that occur after peripheral injury that underlie heterosynaptic facilitation

(Latremoliere and Woolf, 2009; Ruscheweyh et al., 2011; Woolf, 2011).

Hyperalgesia and allodynia reflect peripheral, spinal, and central dysfunctions in the processing of the low-threshold and nociceptive afferent information, suggesting that this type of plastic change might be important for neurobiological processes associated with persistent pain. Increases and decreases in excitatory or inhibitory amino acids during central sensitization are responsible for the excitability changes in projecting spinal cord cells (Willis, 2002). A proposed long chain of metabolic neurotransmitter cascades, including calcium/calmodulin-dependent kinase II (CaMKII) and upregulated phosphorylation of ionotropic glutamate receptors after central sensitization and comparable events, could be achieved during LTP in brain structures (Willis, 2002). It is also important to consider that increases in pain sensitivity involve spinal thalamocortical pathways (Ikeda et al., 2003).

The HFS stimulation was associated with high-frequency and long-lasting postdischarges, which continued throughout the experiment in the form of a burst, possibly of epileptic-like activity (Yamashiro et al.,

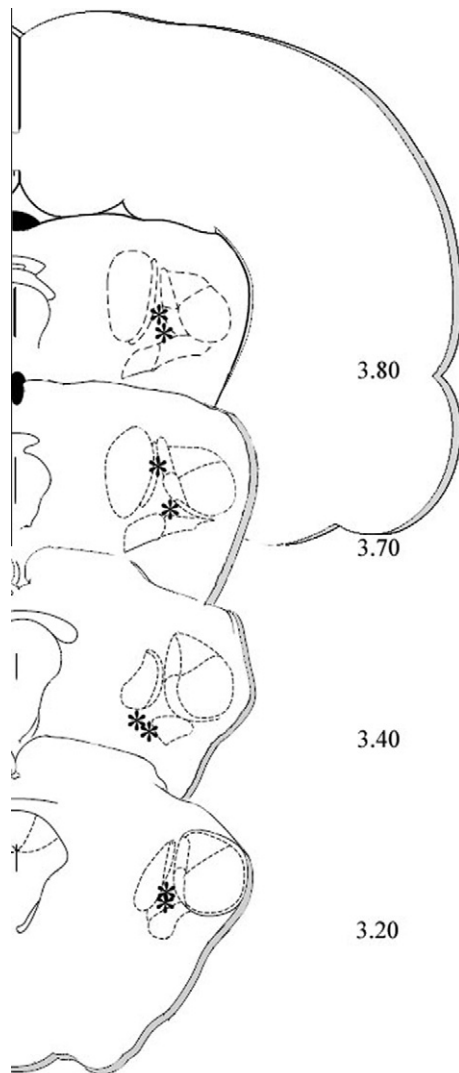


Fig. 9. Drawing taken from the Paxinos and Watson atlas showing the location of the recorded PoT cells.

1991, 1994, 1997; Tremont-Lukats et al., 2000; Waszkielewicz et al., 2011). This burst of epileptic-like activity could be at the neurological basis of persistent pain and pain sensitization processes (Lenz et al., 1989). Moreover, an important relation between pain and epilepsy has been established by the use of anticonvulsant drugs during pain processes (Backonja, 2000; Tremont-Lukats et al., 2000; Hussain and Afshan, 2008). However, this point as well as the definition and extension of receptive fields as a consequence of LTP need to be further explored.

CONCLUSION

Taken together, our results suggest that spinal LTP produces trans-synaptic changes, unmasking the particular properties of the PoT importance for nociceptive recognition. These characteristics make the PoT an interesting region in which to study noxious processing in long-lasting pain process.

CONFLICT OF INTEREST

The authors state that they have no conflict of interest.

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